

JULIEN RABAULT

AI Engineer | ML Engineer

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4 years at CNRS training and fine-tuning deep learning models in production (PyTorch, multi-GPU, Jean Zay supercomputer), 2 peer-reviewed publications. Currently at Berger-Levrault: designing Athena, a multi-agent agentic platform (LangGraph, RAG, MCP).

TECHNICAL SKILLS

Agentic & RAG :	RAG / GraphRAG, Multi-agent systems, MCP Protocol, LangChain / LangGraph, Prompt Engineering, Structured Outputs, Embeddings, LLM Routing, Fine-tuning
Deep Learning :	PyTorch, Transformers, Computer Vision, Diffusion Models (DDPM), NLP, CNNs / U-Net / YOLOv5, Generative models (GAN, VAE)
MLOps & Infra :	Docker, AWS, CI/CD, MLFlow, Airflow, Kubernetes, Celery, Langfuse, HPC / Slurm, Linux
Development :	Python, FastAPI, HuggingFace, Qdrant / pgvector, Mistral / OpenAI API, Git, SOLID / Architecture, C#, SQL

WORK EXPERIENCE

AI ENGINEER Jan. 2026 - present

Berger-Levrault | Toulouse | AI R&D team, 12 people

Designing and building **Athena**, Berger-Levrault's agentic platform: multi-agent orchestration, intelligent routing, connected to documents and business APIs across all client verticals (local government, industry, maintenance). Currently deployed to ~30 pilot users.

Multi-agent architecture

Designed the multi-agent architecture (LangGraph), built question-type routing, orchestrated RAG agents and MCP API agents, integrated source attribution into responses. **Platform in production**, delivering grounded and actionable answers across business domains.

Content extractor - OCR/PDF/DOCX

Rebuilt the extraction service from scratch: OCR, images, PDF, DOCX. Async batch processing (Celery + Mistral batch API), factory/registry pattern for easy format extension. **Halved extraction costs**.

Airflow pipelines

Took over and improved document ingestion pipelines (PDF, technical manuals, work orders, machine docs). **5 operational DAGs** (one per client).

MCP Builder

An LLM pre-processes OpenAPI specs (endpoint grouping, masking, description generation), followed by human-in-the-loop review to validate routes, docs and domain knowledge. **120+ internal APIs mapped**, progressively integrated at runtime.

MACHINE LEARNING ENGINEER Dec. 2021 - Jan. 2026

CNRS - National AI Research Programme (PNRIA) | Toulouse

Worked with research teams across France on applied AI challenges. Ran 2 projects in parallel (6-12 months each), delivered to Meteo France, CNES, CEA, INEE. Training and fine-tuning on Jean Zay (multi-GPU DDP, up to 8 GPUs, Slurm). PyTorch profiling.

GENS / MetScore - Meteo France

Evaluated weather models in production, multi-GPU optimisation and fine-tuning of a diffusion model (DDPM) in PyTorch on Jean Zay. Built MetScore (YAML config, Python library). **Library still in production** + diffusion POC achieving **20% compute savings** at equal quality. Co-author, AMS 2025 paper.

DeepFaune - CNRS/INEE

Fine-tuned YOLOv5 on a custom dataset (1.5M images, 24 classes), multi-GPU training, class imbalance handling, precision/speed optimisation for CPU deployment. **93% accuracy across 24 species, 3x faster**. Peer-reviewed publication.

Other contributions

AUTOFILL (CEA, PairVAE, nanomaterial data generation, MAE 0.98), BIGSF (CNES, tech lead and architecture refactor, galactic filament image analysis library), MORPHOGAN (Univ. Lorraine, StyleGAN2 code overhaul, automated pipeline).
Course delivered: *Introduction to LLMs* (3 hrs, ~25 PhD students and CNRS researchers).

SOFTWARE ENGINEER (apprenticeship) Aug. 2020 - Sept. 2021

Agileo Automation | Montauban

Supervision and control framework for robotic machinery in semiconductor manufacturing. C#, object-oriented architecture, HMI, CI/CD. Team of 5 engineers, Agile/Scrum.

EDUCATION

MSc - ARTIFICIAL INTELLIGENCE & PATTERN RECOGNITION (IARF) Universite Paul Sabatier Toulouse III - IRIT Deep Learning, Computer Vision, NLP	2019 - 2021
BSc - COMPUTER SCIENCE Universite Paul Sabatier Toulouse III	2016 - 2019
Languages: French (native) English (professional working proficiency, scientific writing)	

PUBLICATIONS

"Enriching Operational High-Resolution Ensemble Forecasts with StyleGAN-2" - AIES, 2025. **3rd author**, peer-reviewed.
"The DeepFaune initiative: automatic identification of European fauna" - **Julien Rabault** et al. **Co-author**, peer-reviewed.

OPEN SOURCE PROJECTS

LLMock (PyPI) - LLM mock server for testing retries, fallbacks and rate limiting. Python, FastAPI, 10+ providers, OpenAI-compatible.
DDPM-weather - Probabilistic diffusion model for weather image denoising, 20% resource savings. PyTorch, Meteo France.
BananaML - End-to-end ML pipeline deployed on AWS. Computer Vision + REST API. FastAPI, Docker, CI/CD.